

DGLF: A Dual Graph-based Learning Framework for Multi-modal Sarcasm Detection

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Abstract

Capturing inter-modal incongruities within the text-image pair is a critical challenge in multi-modal sarcasm detection (MSD). Fortunately, graph neural networks (GNNs) have made promising advancements in MSD, which show advantages in explicitly capturing data relationships. Nevertheless, current GNN-based MSD methods do not effectively address some of the inherent limitations of GNNs, which include: 1) neglecting high-order relationships, and 2) underestimating high-frequency messages. In this paper, we propose a **Dual Graph-based Learning Framework (DGLF)** to address the above two issues. Specifically, we construct a hypergraph to perform high-order aware propagation and a vanilla graph to perform high-frequency enhanced propagation, respectively. We empower GNNs to 1) better capture the inherent and complicated relationships based on the hypergraph and 2) deliver sufficient modeling through high-frequency enhanced messages on the vanilla graph. Besides, we introduce multi-modal fusion information bottleneck to effectively fuse the two learned graph features. Experimental results on two benchmark datasets show that the proposed model outperforms previous state-of-the-art methods.

1 Introduction

Due to the rise of social media platforms such as X and Facebook, multi-modal sarcasm detection (MSD) has garnered increasing research attention. MSD aims to recognize the sarcastic sentiment in multi-modal social posts (Cai et al., 2019), which typically refer to textual sentences accompanying images. Unlike traditional text-only sarcasm detection (Riloff et al., 2013; Poria et al., 2016; Zhang et al., 2016) focusing on inconsistencies in expression within the text, the key objective of MSD is



(a) what a gorgeous day (b) feeding my abs nothing but the best quality beef

Figure 1: Two examples of multi-modal sarcasm. In example (a), the text refers to a “gorgeous” day, but the accompanying image shows heavy rain, indicating sarcasm. In example (b), the text mentions “the best quality beef”, but the image displays fast food beef burgers, suggesting sarcasm as well.

to effectively identify subtle inter-modal inconsistencies in the expression of sentiment within an image-text pair, as shown in Figure 1.

Towards this goal, a group of MSD works attempt to concatenate the textual and visual features to encapsulate sarcastic information (Schifanella et al., 2016), or leverage attention mechanism (Vaswani et al., 2017) to *implicitly* fuse features across modalities based on external knowledge (Cai et al., 2019; Pan et al., 2020). More recently, Graph Neural Networks (GNNs) have achieved remarkable advancements in MSD, showcasing their exceptional ability by *explicitly* extracting structural information (Liang et al., 2021, 2022; Liu et al., 2022). As shown in Figure 2(a), the conventional approach in this paradigm constructs a heterogeneous graph, where each token from both modalities is treated as a node, with similarity-based edge construction or carefully adjusted edge weighting strategies. On this basis, it enables simultaneous modeling of inter- and intra-modal token dependencies through message passing, facilitating tighter entanglement and richer interactions.

Despite the promising progress these GNN-based MSD models have achieved, we discover

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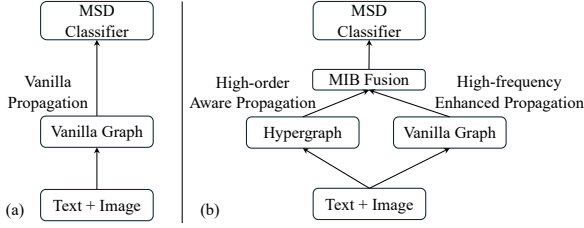


Figure 2: Conceptual comparison of the state-of-the-art methods (a) and the proposed model DGLF (b). “MIB” is short for multi-modal information bottleneck.

that they still suffer from two main issues:

(1) **Neglect of high-order relationship exploration.** They predominantly conceptualize relationships among tokens in pairwise formulations, providing merely an approximation of higher-order relationships through the aggregation of multiple pairs (Feng et al., 2019; Bai et al., 2021). As depicted in Figure 1(b), crucial visual information “*burger*”, which aligns with sarcastic textual cues “*best quality beef*”, may be scattered across the image (two burgers within the image). While it is feasible to construct edges simultaneously among two visual tokens involving “*burger*” and textual tokens of “*best quality beef*”, simplifying these high-order relationships into pairwise formulation might compromise their expressiveness (Sun et al., 2021). Therefore, the sophisticated and nuanced high-order relationships may not be fully captured by existing GNN-based MSD methods.

(2) **Overlooking high-frequency messages exploitation.** The propagation rule of GNNs, characterized by the aggregation and smoothing of messages from neighboring nodes, is widely regarded as an analogy to a fixed low-pass filter (Wu et al., 2019). It predominantly facilitates the flow of low-frequency messages in the graph while significantly attenuating high-frequency messages (Bo et al., 2021). Conversely, in GNNs for MSD, the high-frequency messages may be more vital which reflects discrepancies and inconsistencies in the expression of sentiment. As such, the potential of high-frequency information remains largely unexploited in existing GNN-based MSD frameworks.

To tackle the above issues, as shown in Figure 2(b), we propose a **Dual Graph-based Learning Framework** termed DGLF for MSD. **For the first issue**, we construct a hypergraph (Feng et al., 2019) with edge-dependent node weights (Chitra and Raphael, 2019) to facilitate high-order aware propagation, where each token from both modalities is

represented as a node. We construct intra- and inter-modal hyperedges, which can connect an arbitrary number of nodes. In this fashion, DGLF enables the natural encoding of high-order relationships beyond pairwise formulation. **For the second issue**, we construct another vanilla graph to perform high-frequency enhanced propagation, by adopting a set of frequency filters (Dong et al., 2021; Bo et al., 2021), which distill different frequency constituents from node features. By adaptively integrating high-frequency enhanced messages, DGLF effectively captures sarcastic inconsistencies in local neighborhoods, which is vital for MSD. **Moreover**, we introduce multi-modal information bottleneck (Wu et al., 2023; Zhu et al., 2024a) to effectively fuse the learned graph features, which narrows down the solution space, driving the model’s gaze toward shared modality information.

Overall, our contributions are three-fold: (1) We propose DGLF, a novel dual graph-based learning framework for MSD. To our best knowledge, we are the first to introduce hypergraph into MSD. (2) We construct a hypergraph and a vanilla graph to perform high-order aware and high-frequency enhanced propagation, respectively. Besides, we introduce multi-modal information bottleneck to effectively fuse the learned graph features. (3) Extensive experiments show that our model achieves new state-of-the-art results, further analyses confirm the effect of each component of our model.

2 Methodology

Task Definition. Given a sample S_i from the training set, the objective of MSD is to determine whether the sample implies any sarcasm by learning a model $f(\cdot)$ using the text T_i and corresponding image V_i of S_i . This conventional training procedure is represented as $\hat{y}_i = f(T_i, V_i | \Theta) \in \{0, 1\}$, where $\hat{y}_i = 1$ indicates the sample is predicted to be sarcastic and vice versa; Θ represents the learnable model parameters. For simplicity, we temporally omit the superscript i that indexes the training samples in the following.

Next, before diving into the details of the proposed DGLF’s architecture, we first introduce the feature encoding (§2.1) of modalities and construction of the dual graphs (§2.2) for the MSD task.

2.1 Feature Encoding

For a fair comparison with previous works, given a textual sentence $T = \{t_1, t_2, \dots, t_N\}$ consist-

ing of N words, we adopt the pre-trained BERT model (Devlin et al., 2019), to map each word t_* into d -dimensional embedding,¹ denoted as $\mathbf{H}^t \in \mathbb{R}^{N \times d}$. For a given image $\mathbf{V} \in \mathbb{R}^{L_h \times L_w}$, following Liang et al. (2021); Liu et al. (2022), we first resize it to 224×224 pixels, *i.e.*, $L = L_h = L_w = 224$. Then the image is divided into $M = p \times p$ patches², *w.r.t.* $\mathbf{V} \in \mathbb{R}^{M \times (L/p \times L/p)}$. Subsequently, we feed the sequence of M image patches into a Vision Transformer (ViT) (Dosovitskiy et al., 2021) with an MLP layer to acquire the visual representation $\mathbf{H}^v \in \mathbb{R}^{M \times d}$.

2.2 Graph Construction

2.2.1 Hypergraph Construction

After we obtain textual and visual representations $\mathbf{H}_t$ and $\mathbf{H}_v$ through §2.1, we construct a hypergraph $\mathcal{G}$ with edge-dependent node weights (Feng et al., 2019; Chitra and Raphael, 2019) from the representations as shown in Figure 3(a, b).

Mathematically, denote a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \tau, \zeta)$ in which each node $v_i^\delta \in \mathcal{V}$ ($\delta \in \{t, v\}, |\mathcal{V}| = N + M$) corresponds to a textual token v_i^t or visual token v_i^v , where we initialize the node embeddings $\{v_i^t, v_i^v\}$ with the encoded representations $\{h_i^t, h_i^v\}$, respectively.

For every hyperedge $e \in \mathcal{E}$ ($|\mathcal{E}| = 2 + N + M$), it encodes intra- or inter-modal dependencies. Specifically, each node v_i^δ ($\delta \in \{t, v\}$), where i spans the range 1 to N_δ with N_δ denoting the total number of tokens in modality δ , is first connected to all other tokens in the same modality $\{v_j^\delta | j \neq i, 1 \leq j \leq N_\delta\}$ through a single intra-modal hyperedge. Here we obtain 2 intra-modal hyperedges. Furthermore, each node v_i^δ is connected to all tokens in the opposite modality $\{v_k^{\bar{\delta}} | 1 \leq k \leq N_{\bar{\delta}}\}$, with $\bar{\delta}$ indicating the modality opposite to δ via an inter-modal hyperedge, where $N_{\bar{\delta}}$ represents the total number of tokens in modality $\bar{\delta}$. Here we obtain N and M inter-modal hyperedges from N textual tokens and visual tokens, respectively. The sum of $N + M$ inter-modal hyperedges and 2 intra-modal hyperedges results in $N + M + 2$ hyperedges.

Unlike prior methods (Liang et al., 2021, 2022; Liu et al., 2022) that resort to adjustments of edge weighting strategies through complex relationship learning or similarity metrics, our DGLF embraces simplicity by adopting randomly initialized weight

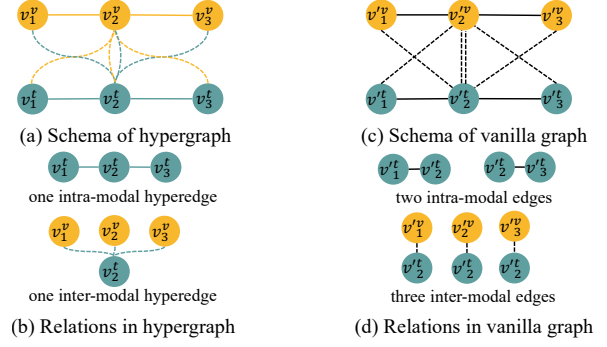


Figure 3: The illustration of hypergraph (a, b) and vanilla graph (c, d), *w.l.o.g.*, only the edges directed into v_2^v , v_2^t , v_2^v and v_2^t are shown. As demonstrated, the hypergraph is adept at modeling high-order relationships, where four vertices in (a) (*e.g.*, v_1^v , v_2^v , v_3^v and v_2^t) are connected by a single hyperedge sharing the same color green; and in (c), the edge construction adheres to the pairwise relationship assumption.

values. Concretely, we introduce two distinct categories of weights in $\mathcal{G}$: (1) an edge weight $\tau(e)$ assigned to each hyperedge e , and (2) a node weight $\zeta_e(v)$ for every node v upon which hyperedge e is incident, denoted as edge-dependent node weight (Chitra and Raphael, 2019). $\zeta_e(v)$ quantifies the significance of node v within hyperedge e , thereby reinforcing fine-grained intra- and inter-modal relationships. Denote $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$ represent the incidence matrix, in which a nonzero entry $\mathbf{A}_{ve} = 1$ indicates that the hyperedge is incident with the node v ; otherwise $\mathbf{A}_{ve} = 0$. Formally, edge-dependent node weights can be represented by a weighted incidence matrix $\hat{\mathbf{A}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$:

$$\hat{\mathbf{A}} = \begin{cases} \zeta_e(v), & \text{if } e \text{ is incident with node } v, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

2.2.2 Vanilla Graph Construction

As previously discussed, high-frequency information that reflects emotional discrepancies may be more pivotal for MSD (Bo et al., 2021; Wu et al., 2019), and combining the power of messages with different frequencies is worth exploring. This insight compels us to introduce a high-frequency enhanced propagation aimed at distilling varying frequency importance. To this end, we further construct a vanilla graph $\mathcal{G}' = \{\mathcal{V}', \mathcal{E}'\}$ from the multi-modal encoded representations as shown in Figure 3(c, d), in parallel with the hypergraph. Mathematically, denote a vanilla graph $\mathcal{G}' = \{\mathcal{V}', \mathcal{E}'\}$ whose nodes $\mathcal{V}'$ are identical to the ones in $\mathcal{G}$, denoted with $\{v_i^t, v_j^v\}$. The node embeddings are similarly initialized with the encoded

¹Following Liang et al. (2021, 2022), we adopt the first sub-token's representation as the whole word representation.

²In line with previous works, p is set to 7.

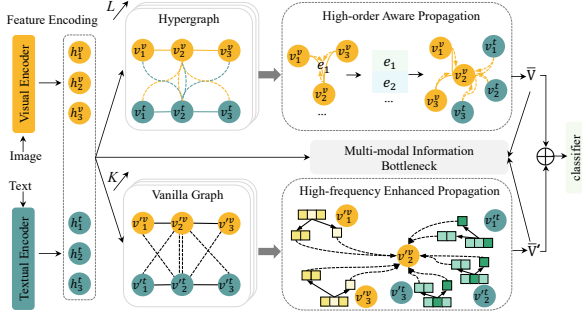


Figure 4: The architecture of DGLF, where dotted directed arrows in both propagation modules delineate the direction of message passing. Note that in High-frequency Enhanced Propagation component, $\blacksquare$ and $\blacksquare$ represent visual and textual high-frequency messages, while $\square$ and $\square$ represent visual and textual low-frequency messages. (Zoom-in for better view)

representations $\{h_i^t, h_j^v\}$ as well. The normalized graph Laplacian matrix can be represented as $\mathbf{L} = \mathbf{I} - \mathbf{D}_{\mathcal{G}'}^{-1/2} \mathbf{A}' \mathbf{D}_{\mathcal{G}'}^{-1/2}$, where $\mathbf{A}' \in \mathbb{R}^{|\mathcal{V}'| \times |\mathcal{V}'|}$ denotes the adjacency matrix, $\mathbf{D}_{\mathcal{G}'}$ denotes a diagonal degree matrix and $\mathbf{I}$ denotes an identity matrix.

2.3 Model Architecture

This section introduces the details of our proposed DGLF, whose architecture is shown in Figure 4.

2.3.1 High-order Aware Propagation

The main objective of constructing the hypergraph is to explore the subtle high-order sarcastic information within and across modalities. Concretely, we first conduct node convolution by aggregating node features to update hyperedge embeddings, and then conduct hyperedge convolution to spread the hyperedge messages to the nodes:

$$\mathbf{V}^{(\ell+1)} = \sigma(\mathbf{D}_{\mathcal{G}}^{-1} \mathbf{A} \mathbf{W}_e \mathbf{B}^{-1} \hat{\mathbf{A}}^\top \mathbf{V}^{(\ell)}), \quad (2)$$

in which $\mathbf{V}^{(\ell)} = \{v_{i,(\ell)}^\delta | \delta \in \{t, v\}, i \in [1, N] \text{ when } \delta = t; i \in [1, M] \text{ when } \delta = v\}$, $\mathbf{V}^{(\ell)} \in \mathbb{R}^{|\mathcal{V}| \times d}$ is the input at layer ℓ ; σ is a non-linear activation function; $\mathbf{W}_e = \text{diag}(\tau(e_1), \dots, \tau(e_{|\mathcal{E}|}))$ denotes the hyperedge weight matrix; $\mathbf{D}_{\mathcal{G}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and $\mathbf{B} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{E}|}$ denote the node degree and hyperedge degree matrix. In this manner, the high-order inter- and intra-modal relationships are gradually refined. After L iterations, we get the outputs of the last iteration $v_{i,(L)}^\delta$ as the high-order aware representations.

2.3.2 High-frequency Enhanced Propagation

To propagate high-frequency enhanced information on the vanilla graph, we first design a low-pass filter

$\mathcal{F}_l$ and a high-pass filter $\mathcal{F}_h$ to distill the messages from the node features:

$$\begin{aligned} \mathcal{F}_l &= \mathbf{I} + \mathbf{D}_{\mathcal{G}'}^{-1/2} \mathbf{A}' \mathbf{D}_{\mathcal{G}'}^{-1/2} = 2\mathbf{I} - \mathbf{L}, \\ \mathcal{F}_h &= \mathbf{I} - \mathbf{D}_{\mathcal{G}'}^{-1/2} \mathbf{A}' \mathbf{D}_{\mathcal{G}'}^{-1/2} = \mathbf{L}. \end{aligned} \quad (3)$$

Note that $\mathcal{F}_h$ is equivalent to the normalized graph Laplacian matrix, which is consistent with the theory that the Laplacian kernel can be employed to highlight high-frequency information (Jain and Farrokhnia, 1991).

Now, we employ these two filters to adaptively aggregate messages with varying frequencies. To be specific, we use a weighted sum to combine low-frequency and high-frequency messages:

$$\begin{aligned} \mathbf{V}'^{(k+1)} &= \mathbf{W}^l (\mathcal{F}_l \cdot \mathbf{V}'^{(k)}) + \mathbf{W}^h (\mathcal{F}_h \cdot \mathbf{V}'^{(k)}) \\ &= \mathbf{V}'^{(k)} + (\mathbf{W}^l - \mathbf{W}^h) \mathbf{D}_{\mathcal{G}'}^{-1/2} \mathbf{A}' \mathbf{D}_{\mathcal{G}'}^{-1/2} \mathbf{V}'^{(k)}, \end{aligned} \quad (4)$$

where $\mathbf{V}'^{(k)} = \{v_{i,(k)}^\delta | i \in [1, N] \text{ when } \delta = t; i \in [1, M] \text{ when } \delta = v, \delta \in \{t, v\}\} \in \mathbb{R}^{|\mathcal{V}'| \times d}$ is the input at layer k ; $\mathbf{W}^l, \mathbf{W}^h \in \mathbb{R}^{|\mathcal{V}'| \times |\mathcal{V}'|}$ denote the weight matrices for low- and high-frequency information. It can be further rewritten as:

$$v'_{i,(k)+1} = v'_{i,(k)} + \sum_{j \in \mathcal{N}_i} \frac{w_{ij}^l - w_{ij}^h}{\sqrt{|\mathcal{N}_j|} \sqrt{|\mathcal{N}_i|}} v'_{j,(k)}, \quad (5)$$

where $\mathcal{N}_i$ denotes the neighboring nodes of node i ; w_{ij}^l and w_{ij}^h denote the weight contributions of node j 's low-frequency and high-frequency messages to node i , respectively with $w_{ij}^l + w_{ij}^h = 1$. And they are calculated using a self-gating mechanism similar to Bo et al. (2021).

By stacking K layers, each node receives the high-frequency enhanced messages, which are ignored by previous works, and we utilize outputs of the final layer $v_{i,(K)}^\delta$ as the high-frequency enhanced representations.

2.3.3 Information Bottleneck based Multi-modal Fusion

Ideally, concatenated representations should encapsulate information shared across modalities. Thus, we introduce Multi-modal Fusion Information Bottleneck (MFB) (Wu et al., 2023; Mai et al., 2022), which effectively constrains the solution space to focus more on the shared multi-modal information. Concretely, we first fuse different modalities to obtain concatenated high-order aware and high-frequency enhanced representations as follows:

$$\bar{\mathbf{V}} = \mathbf{V}_{(L)}^t \oplus \mathbf{V}_{(L)}^v, \quad \bar{\mathbf{V}}' = \mathbf{V}_{(K)}^t \oplus \mathbf{V}_{(K)}^v, \quad (6)$$

Denote $\mathbf{H} = \mathbf{H}_t \oplus \mathbf{H}_v$, the MFB for concatenated high-order aware and high-frequency enhanced representations are calculated as follows:

$$\begin{aligned} \min_{p(\bar{\mathbf{V}}|\mathbf{H}) \in \Omega_1} \text{MFB}(\bar{\mathbf{V}}; \mathbf{H}) &\triangleq - \sum_{\delta} \mathcal{I}(\mathbf{H}_{\delta}; \bar{\mathbf{V}}) + \beta \mathcal{I}(\mathbf{H}; \bar{\mathbf{V}}), \\ \min_{p(\bar{\mathbf{V}}'|\mathbf{H}) \in \Omega_2} \text{MFB}(\bar{\mathbf{V}}'; \mathbf{H}) &\triangleq - \sum_{\delta} \mathcal{I}(\mathbf{H}_{\delta}; \bar{\mathbf{V}}') + \beta \mathcal{I}(\mathbf{H}; \bar{\mathbf{V}}'), \end{aligned} \quad (7)$$

where $\delta \in \{t, v\}$; β serves as a trade-off hyper-parameter; Ω_1 and Ω_2 denote the search space of the conditional distribution of $\bar{\mathbf{V}}$ and $\bar{\mathbf{V}}'$ given the initial concatenated vertex feature $\mathbf{H}$, respectively. We then perform bound estimations to enable its computation and training via back propagation:

Proposition 1. *The upper and lower bounds of mutual information between two random variables $\mathbf{x}$ and $\mathbf{y}$ can be estimated as:*

$$\mathbb{E} \left[\log \frac{f(\mathbf{y}_+, \mathbf{x})}{\sum_{\mathbf{y}_i \in Y} f(\mathbf{y}_i, \mathbf{x})} \right] \leq \mathcal{I}(\mathbf{x}; \mathbf{y}) \leq D_{KL}(p(\mathbf{x}|\mathbf{y})||q(\mathbf{x})), \quad (8)$$

where $\mathbf{x}$ and $\mathbf{y}_+$ are positive pairs sampled from $p(\mathbf{x}|\mathbf{y})$, $f(\cdot, \cdot)$ represents a scoring function that measures the similarity between two embeddings, and q represents a prior distribution of $\mathbf{x}$.

The form of mutual information’s lower bound above is known as the InfoNCE loss (Oord et al., 2018). Thus, the MFB loss can be expressed as :

$$\begin{aligned} \mathcal{L}_{\text{MFB}} &= \sum_{\delta} \mathcal{L}_{\text{InfoNCE}}(\mathbf{H}_{\delta}, \bar{\mathbf{V}}) + \beta D_{KL}(p(\bar{\mathbf{V}}|\mathbf{H})||q(\bar{\mathbf{V}})) \\ &+ \sum_{\delta} \mathcal{L}_{\text{InfoNCE}}(\mathbf{H}_{\delta}, \bar{\mathbf{V}}') + \beta D_{KL}(p(\bar{\mathbf{V}}'|\mathbf{H})||q(\bar{\mathbf{V}}')). \end{aligned} \quad (9)$$

2.4 Training Objective

After the above procedures, we follow Liang et al. (2021, 2022) to employ attention mechanism (Vaswani et al., 2017) based on $\mathbf{H}$ and $\mathbf{V}$ (resp. $\mathbf{V}'$) to obtain the high-order aware presentation $\mathbf{f}_1$ (resp. high-frequency enhanced presentation $\mathbf{f}_2$). Now, we concatenate $\mathbf{f}_1$ and $\mathbf{f}_2$ to obtain the final representation $\mathbf{f}$, which is then fed into a fully-connected layer with softmax normalization to capture a probability distribution $\hat{\mathbf{y}} \in \mathbb{R}^{d_p}$ of sarcasm detection space as follows:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_o \mathbf{f} + \mathbf{b}_o), \quad (10)$$

where d_p is the dimensionality of sarcasm labels; $\mathbf{W}_o$ and $\mathbf{b}_o \in \mathbb{R}^{d_p}$ are trainable parameters. For MSD task, we minimize the cross-entropy loss via the standard gradient descent algorithm:

$$\mathcal{L}_{\text{MSD}} = - \sum_{i=1}^N \sum_{j=1}^{d_p} \mathbf{y}_i^j \log \hat{\mathbf{y}}_i^j + \lambda ||\Theta||^2, \quad (11)$$

where N denotes the training data size; $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ represent the ground-truth and predicted label distribution of instance i , respectively; Θ denotes all trainable parameters of our DGLF and λ denotes the coefficient of L_2 -regularization.

The final training objective is the sum of $\mathcal{L}_{\text{MSD}}$ (Eqn. (11)) and $\mathcal{L}_{\text{MFB}}$ (Eqn. (9)).

3 Experiments

3.1 Main Results

Due to space limitation, we put experiment settings in Appendix A.1. The performance comparison of our model and baselines are shown in Table 1, from which we have the following observations:

(1) **Our model achieves new state-of-the-art (SOTA) performance on all metrics and datasets.**

Specifically, on MMSD, DGLF overpasses HKE by 1.62% and 2.91% on Acc. and F1; on MMSD2.0, it overpasses Att-BERT by 1.49% and 1.56% on Acc. and F1 respectively. This is because our model captures complicated high-order relationships based on the hypergraph, and our designed high-frequency enhanced propagation based on the vanilla graph further improves the model’s ability to detect inconsistencies in sarcasm.

(2) **The improvements on MMSD dataset are much sharper.**

This can be attributed to the fact that MMSD2.0 eliminated obvious sarcastic cues in MMSD, which has led to an obvious performance decline in existing GNN-based methods that rely on edge weighting strategies with complex relationship learning or similarity metrics, sometimes even underperforming compared to Att-BERT. Thanks to the high-order and high-frequency information captured, our model achieves consistent improvements over all baselines on both datasets.

(3) **Based on more advanced feature encoders (e.g., CLIP), our DGLF can still achieve significant improvements.**

We suspect the reason is that the advantages of our approach are orthogonal to the ability of feature encoders. Our method can teach CLIP to model high-order relationships and high-frequency messages on dual graphs, which can hardly be learned in the pre-training process.

3.2 Method Analysis

Effect of High-order Aware Propagation. One of the core contributions of our work is modeling the high-order relationships based on the hypergraph, while previous GNN-based works only adhere to the pairwise relationship formulation. To

Modality	Model	MMSD				MMSD2.0			
		Acc. (%)	P (%)	R (%)	F1 (%)	Acc. (%)	P (%)	R (%)	F1 (%)
Text-only	TextCNN (Kim, 2014)	80.03	74.29	76.39	75.32	71.61	64.62	75.22	69.52
	Bi-LSTM (Graves and Schmidhuber, 2005)	81.90	76.66	78.42	77.53	72.48	68.02	68.08	68.05
	SMSD (Xiong et al., 2019)	80.90	76.46	75.18	75.82	75.36	68.45	71.55	69.97
	BERT* (Devlin et al., 2019)	83.60	78.50	82.51	80.45	76.52	74.48	73.09	73.78
Image-only	ResNet (He et al., 2016)	64.76	54.41	70.80	61.53	65.50	61.17	54.39	57.58
	ViT* (Dosovitskiy et al., 2021)	68.51	57.19	70.83	63.28	71.80	64.96	75.15	69.68
Multi-modal	HFM (Cai et al., 2019)	83.44	76.57	84.15	80.18	70.57	64.84	69.05	66.88
	D&R Net (Xu et al., 2020)	84.02	77.97	83.42	80.60	-	-	-	-
	Att-BERT (Pan et al., 2020)	86.05	80.87	85.08	82.92	80.03	76.28	77.82	77.04
	<i>GNN-based, BERT & ViT as Encoder</i>								
	InCrossMGs (Liang et al., 2021)	86.10	81.38	84.36	82.84	-	-	-	-
	CMGCN (Liang et al., 2022)	86.54	-	-	82.73	79.83	75.82	78.01	76.90
	HKE* (Liu et al., 2022)	87.39	81.40	86.93	84.07	76.39	73.50	75.96	74.71
	DGLF (Ours)	89.01[†]	84.96[†]	89.10[†]	86.98[†]	81.52[†]	77.98[†]	79.23[†]	78.60[†]
	<i>CLIP as Encoder</i>								
	Multi-view CLIP* (Qin et al., 2023)	88.22	82.03	88.13	84.97	85.14	80.18	88.21	84.00
	DGLF _{CLIP} (Ours)	89.43[†]	85.81[†]	89.27[†]	87.51[†]	86.82[†]	81.90[†]	89.85[†]	85.69[†]

Table 1: Results comparison. * denotes our re-implementation using the official code. - denotes missing results from the published work. Since CLIP-based methods use different pre-trained feature encoders, we gray out them for a fair comparison. [†] denotes the significance tests of DGLF and DGLF_{CLIP} over baselines at p -value < 0.05 .

verify its effect, we design variant 1 as shown in Table 2. We can observe that Acc. drops by 1.95% on MMSD and 1.65% on MMSD2.0. Moreover, F1 drops more significantly: 2.16% on MMSD and 1.81% on MMSD2.0. This proves that modeling the high-order relationships on the hypergraph $\mathcal{G}$ can naturally encode higher arity relationships within and between visual and textual elements, thus significantly improving sarcasm detection.

In §2.2.1, we define two types of weights in hypergraph $\mathcal{G}$ to capture the high-order relationships at a fine-grained level. To study its effect, we conduct the ablation experiments by removing different sets of them. It can be seen that from variant 2 to 4 in Table 2 that removing either or both (*i.e.*, setting weight value $\tau(e)$ or/and $\zeta_e(v)$ as 1) leads to performance decreases in all metrics on both datasets. This indicates that the formulated weights in hypergraph $\mathcal{G}$ benefit the final performance.

Effect of High-frequency Enhanced Propagation. The aim of constructing another vanilla graph $\mathcal{G}'$ is to perform high-frequency enhanced propagation. To verify its effectiveness, we design variant 5 by performing predictions using the high-order aware propagation on the hypergraph only. From Table 2, we find that variant 5 obtains dramatic drops in all metrics on both datasets. The ap-

parent performance gap verifies the high-frequency enhanced messages in MSD, which can capture the varying importance of sentiment discrepancy and commonality within local neighborhoods.

To further analyze the propagation of only high-frequency and low-frequency information in the vanilla graph $\mathcal{G}'$, we set $\mathbf{R}^l$ and $\mathbf{R}^h$ in Eqn. (4) to zero, respectively. From variant 6 and variant 7 in Table 2, we observe that erasing high-frequency information results in a more noticeable performance decline. This fully demonstrates the necessity of high-frequency information for the MSD task and intuitively supports our motivation.

Effect of Multi-modal Information Bottleneck.

From Table 2 variant 8, we can find that removing $\mathcal{L}_{\text{MFB}}$ in final training loss leads to large performance decreases. Specifically, Acc. drops 1.28% and 0.56% on MMSD and MMSD2.0; F1 drops 1.37% and 0.67% on MMSD and MMSD2.0. This can verify the effectiveness of MFB, which constrains the solution space of the two learned graph representations to focus on cross-modal shared information, thus boosting performance.

Sensitivity of Hyper-parameter. To investigate the robustness of our proposed multi-modal information bottleneck, we conduct sensitivity analyses.

#	Variant	MMSD		MMSD2.0	
		Acc. (%)	F1 (%)	Acc. (%)	F1 (%)
-	DGLF	89.01	86.98	81.52	78.60
<i>Effect of High-order Aware Propagation</i>					
1	w/o high-order aware propagation	87.06 ($\downarrow 1.95$)	84.82 ($\downarrow 2.16$)	79.87 ($\downarrow 1.65$)	76.79 ($\downarrow 1.81$)
2	w/o edge weight $\tau(e)$	87.25 ($\downarrow 1.76$)	85.17 ($\downarrow 1.81$)	80.81 ($\downarrow 0.71$)	77.80 ($\downarrow 0.80$)
3	w/o node weight $\zeta_e(v)$	87.81 ($\downarrow 1.20$)	85.69 ($\downarrow 1.29$)	80.98 ($\downarrow 0.54$)	77.97 ($\downarrow 0.63$)
4	w/o both weights	87.08 ($\downarrow 1.93$)	84.86 ($\downarrow 2.12$)	80.35 ($\downarrow 1.17$)	77.31 ($\downarrow 1.29$)
<i>Effect of High-frequency Enhanced Propagation</i>					
5	w/o high-frequency enhanced propagation	86.73 ($\downarrow 2.28$)	84.17 ($\downarrow 2.81$)	79.51 ($\downarrow 2.01$)	76.28 ($\downarrow 2.32$)
6	w/o low-frequency messages included	88.07 ($\downarrow 0.94$)	85.86 ($\downarrow 1.12$)	80.67 ($\downarrow 0.85$)	77.59 ($\downarrow 1.01$)
7	w/o high-frequency messages included	87.84 ($\downarrow 1.17$)	85.60 ($\downarrow 1.38$)	80.46 ($\downarrow 1.06$)	77.37 ($\downarrow 1.23$)
<i>Effect of Multi-modal Information Bottleneck</i>					
8	w/o multi-modal information bottleneck	87.73 ($\downarrow 1.28$)	85.61 ($\downarrow 1.37$)	80.96 ($\downarrow 0.56$)	77.93 ($\downarrow 0.67$)

Table 2: Results of ablation experiments.

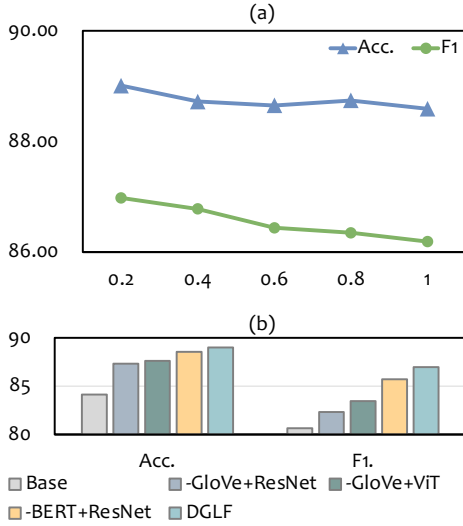


Figure 5: (a) Performance variations when altering the value of β in MFB loss (Eqn. (9)) on MMSD dataset. (b) Performance of using different pre-trained methods on MMSD. **Base** denotes without the proposed dual graphs, *i.e.*, only using BERT and ViT to conduct MSD.

We varied the value of β from 0.2 to 1.0 as per Eqn. (9) and the results are displayed in Figure 5(a). As depicted, the performance remains relatively stable across different values of β , albeit with a slight decreasing trend as β increases. This suggests that our model is largely insensitive to β . However, as β increases and the constraint tightens, there is a gradual effect on the model performance.

Impact of Graph Layers. To investigate the impact of stacking different graph layers for hyper-graph $\mathcal{G}$ and vanilla graph $\mathcal{G}'$, we conduct a grid

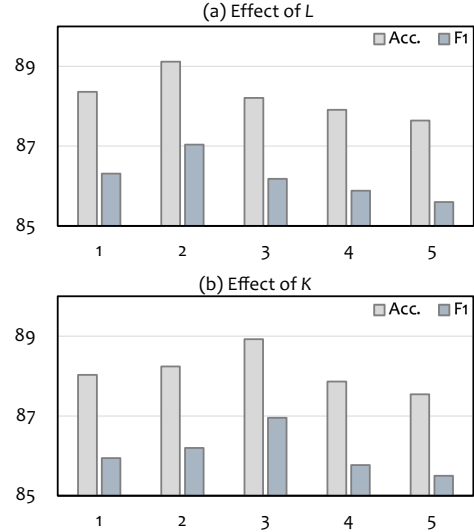


Figure 6: Results of the proposed DGLF at different graph layers (a) L and (b) K on MMSD dataset.

search on the number of layers L and K , respectively. Specifically, we search them from 1 to 5 on the validation set and the results on MMSD dataset are shown in Figure 6. We observe that the effects of L and K are similar. At first, the results steadily improve as stacking more layers, and peak at $L = 2$ and $K = 3$ respectively. Further stacking more layers has little positive impact, as it may incorporate noise from neighborhoods. For MMSD2.0, we empirically found that the performance is not particularly sensitive to the number of layers, displaying no specific pattern.

Generalizability of Dual Graphs. To evaluate the generalizability of the proposed dual graphs

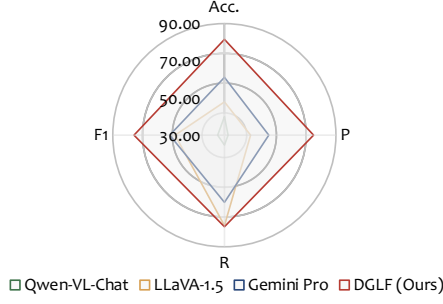


Figure 7: Comparison with large vision-language models (LVLMs) on MMSD2.0 dataset.

Image	Text	(a)	Image	Text	(b)
	nice warm running weather this morning ...			i just love ... work before everyone else, waking up at 3:30 am is so rewarding .	
Model	Prediction		Model	Prediction	
CMGCN	sarcastic	✓	CMGCN	non-sarcastic	✗
HKE	non-sarcastic	✗	HKE	non-sarcastic	✗
DGLF	sarcastic	✓	DGLF	sarcastic	✓

Figure 8: Case study.

with various pre-trained methods beyond CLIP, we conduct experiments using four variants that employ different textual and visual encoders. From results in Figure 5(b), we find that the dual graphs are compatible with various pre-trained models and perform consistently outperform the baseline model without graphs. This verifies the effectiveness and generalizability of the proposed DGLF. Further, it is evident that using more sophisticated pre-trained methods, such as ViT, BERT and CLIP, leads to superior performance.

Comparison with Large Vision-Language Models (LVLMs). To ascertain the competitive edge of our model, we conducted comparative analyses against prevalent LVLMs including Qwen-VL-Chat (Bai et al., 2023), LLaVA-1.5 (Liu et al., 2024), and Gemini Pro (Team et al., 2023) following Wang et al. (2024). The obvious performance gap between ours and LVLMs in Figure 7 underscores the persistent challenges that LVLMs encounter in MSD, despite their advanced zero-shot learning and chain-of-thought capabilities. It emphasizes the need for dedicated efforts in designing effective MSD frameworks. Combining the power of LVLMs (Chen et al., 2024) to detect sarcasm is an interesting direction in future work.

3.3 Case Study

We present two sarcastic cases in Figure 8 to quantitatively verify the effectiveness of our DGLF.

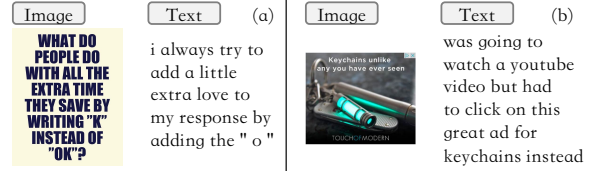


Figure 9: Examples of error prediction.

Model	Acc. (%)	P (%)	R (%)	F1 (%)
DGLF	81.52	77.98	79.23	78.60
+ OCR	81.64	77.10	81.06	79.03

Table 3: Results on incorporating Optical Character Recognition (OCR) information on MMSD2.0 dataset.

Case (a). The sarcasm arises from the contrast between the text, which suggests pleasant and warm conditions, and the image, which displays a much colder temperature typically not considered warm. Despite involving a more complex fusion architecture, HKE still employs a naive GNN propagation framework, transmitting low-frequency messages between nodes. In contrast, our model obtains high-order and high-frequency messages from both the hypergraph and the vanilla graph, and accurately captures the inter-modal inconsistencies, thereby correctly identifying it as sarcastic.

Case (b). The sarcasm likely stems from the exaggeration between the text and the image. The text describes waking up extremely early as “rewarding”, which contrasts sharply with the image accompanying it, depicting the reality of driving early in the morning. Unfortunately, previous SOTA GNN-based methods still predict it as non-sarcastic. Thanks to the proposed dual graph framework, our model can propagate high-frequency messages that reflect discrepancies in the graph, thereby effectively capturing inter-modality inconsistencies and achieving precise prediction.

3.4 Error Analysis

We further conduct error analysis to understand DGLF’s performance. We observe that the majority of errors occur in samples where images contain important textual information, such as the purely textual image in Figure 9 (a) and in Figure 9 (b) which includes both textual and visual expressions. Based on these observations, we conducted a preliminary experiment to leverage textual information within images by integrating OCR into DGLF for MSD. From the results in Table 3, we find a no-

table performance improvement. Thus it would be interesting to effectively leverage important textual information in images for future research.

4 Related Work

Multi-modal Sarcasm Detection. With the rapid popularization of social media platforms, multi-modal sarcasm detection (MSD) has garnered increasing research attention in recent years (Zhu et al., 2024c; Wang et al., 2024). Some early works (Xu et al., 2020; Pan et al., 2020; Xin et al., 2024) focused more on contextual dependencies and utilized feature concatenation for multi-modal modeling. More recently, researchers formulated the MSD task upon GNNs, which have shown promising results. Therein, InCrossMGs (Liang et al., 2021) proposed in-modal and cross-modal graphs to determine the sentiment inconsistencies. Based on this, CMGCN (Liang et al., 2022) explored a cross-modal graph to model the contradictory sentiments between key textual and visual information. HKE (Liu et al., 2022) further introduced a GNN-based hierarchical framework by exploring both the atomic-level congruity and the composition-level congruity.

Nevertheless, these GNN-based MSD models still deliver insufficient high-order relationships and high-frequency messages, as we discussed.

Graph Neural Networks. GNNs can explicitly model data relationships, which have been widely employed in various applications such as sentiment analysis and argument pair extraction (Li et al., 2021; Chen et al., 2023; Sun et al., 2023; Zhu et al., 2024b). GNNs have also inspired MSD researchers and offer a new solution for the MSD task, from unimodal setting (Lou et al., 2021) to multi-modal scenario (Qin et al., 2023).

However, previous works fail to address the general limits of GNNs, which motivates our work.

Multi-modal Information Bottleneck. The InfoMax principle proposed by Linsker (1988) seeks to maximize the mutual information between feature and model output. Along this way, Han et al. (2021) built up a hierarchical mutual information maximization guided model for multi-modal sentiment analysis. Wu et al. (2023) focused on video-based sentiment analysis and used contrastive learning to achieve mutual information maximization.

In this work, we utilize the lower bound of multi-modal information bottleneck (Mai et al., 2022)

to constrain the learned graph features and initial modality features, driving the model’s gaze toward shared modality information.

5 Conclusion

In this paper, we propose a new GNN-based framework for MSD. We construct a hypergraph and a vanilla graph to perform high-order aware propagation and high-frequency enhanced propagation, respectively. Based on this, we introduce multi-modal information bottleneck to effectively fuse the two learned graph representations. Extensive experiments and analyses on two MSD benchmarks show the superiority of our proposed framework.

Limitations

Our DGLF has the following limitations: (1) The proposed dual graph approach (combining hypergraph and vanilla graph) might lead to increased computational complexity. (2) The effectiveness of the DGLF might be contingent upon the quality of the underlying pre-trained models (BERT, ViT), and can benefit from more advanced feature encoders, which is not the focus of this work. (3) DGLF lacks validation on more diverse multi-modal MSD datasets that include additional modalities such as audio and video, as well as testing across various other tasks, which may limit its broader generalizability and effectiveness.

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A Appendix

A.1 Experimental Settings

Datasets. We conduct experiments on two widely-used benchmark datasets: MMSD (Cai et al., 2019) and MMSD2.0 (Qin et al., 2023). Specifically, MMSD is derived from English Twitter. Thereinto, tweets with some special hashtags (e.g., sarcasm) are positive examples and those without such hashtags are negative examples. MMSD2.0 is updated from MMSD, which is the most advanced benchmark in MSD. The statistics of these two datasets are shown in Table 4.

Evaluation Metrics. Following previous works (Liu et al., 2022; Qin et al., 2023), we adopt accuracy (Acc.), precision (P), recall (R), and F1 score (F1) to evaluate the model performance.

Implementation Details. For a fair comparison, we follow Liang et al. (2021, 2022); Liu et al. (2022) to utilize the pre-trained uncased BERT-base model (Devlin et al., 2019) to embed each word as a 768-dimensional embedding and employ the pre-trained ViT (Dosovitskiy et al., 2021) to

MMSD/MMSD2.0	Train	Validation	Test
Sentences	19,816/19,816	2,410/2,410	2,409/2,409
Positive	8,642/9,572	959/1,042	959/1,037
Negative	11,174/10,240	1,451/1,368	1,450/1,372

Table 4: Statistics of two experimental datasets.

embed each visual patch as a 768-dimensional embedding, i.e., $d = 768$. Adam (Kingma and Ba, 2014) is utilized as the optimizer with a learning rate of $2e-5$, and the mini-batch size is 16. The coefficient λ is set to $1e-5$. We test L and K in the range from 1 to 5 on the validation set and choose the best-performing one to the test set, respectively. The hyper-parameter β is set as 0.2. Paired t-test is performed to test the significance of performance improvement with a default significance level of 0.05. All experiments are conducted on one NVIDIA GeForce RTX 3090. The results reported in all experiments are averages of 5 runs with different random seeds to ensure the final reported results are statistically stable.

A.2 Model Zoo

We compared our proposed DGLF with a series of strong baselines, which can be broadly classified into three main categories:

(1) **Text-only methods.** These methods purely rely on textual information for sarcasm detection, including **TextCNN** (Kim, 2014), a deep learning model based on CNN; **Bi-LSTM** (Graves and Schmidhuber, 2005), a bidirectional LSTM network for text classification; **SMSD** (Xiong et al., 2019) explored a self-matching network to capture textual incongruity information; and **BERT** (Devlin et al., 2019), the vanilla pre-trained uncased BERT-base taking ‘[CLS] text [SEP]’ as input. (2) **Image-only methods.** The sarcasm detection in these methods relies solely on image input, including **ResNet** (He et al., 2016) which trains a sarcasm classifier; and **ViT** (Dosovitskiy et al., 2021), which utilizes the ‘[class]’ token representations to detect the sarcasm. (3) **Multi-modal methods.** These methods utilize both visual and textual information for sarcasm detection, including **HFM** (Cai et al., 2019) introduced a hierarchical fusion model for MSD; **D&R Net** (Xu et al., 2020) proposed a decomposition and relation network modeling both cross-modality contrast and semantic association; **Att-BERT** (Pan et al., 2020) explored inter-modality attention and co-attention

to model the incongruity of multimodal information; **InCrossMGs** (Liang et al., 2021) leveraged the sarcasm relations from both intra- and inter-modality perspectives using local multi-modal features; **CMGCN** (Liang et al., 2022) explored the sarcastic relations across objects of the image and tokens of the text; **HKE** (Liu et al., 2022) utilized both the atomic-level congruity based on cross attention and the composition-level congruity based on GNNs; and **Multi-view CLIP** (Qin et al., 2023) employed the pre-trained CLIP (Radford et al., 2021) model to detect different sarcastic cues captured from multiple perspectives.

Further, to investigate the effectiveness of our DGLF when used with different pre-trained models, we also set the following variants: **-GloVe+ViT**: We replace pre-trained BERT in our proposed framework with GloVe (Pennington et al., 2014) to initialize each word into a 300-dimensional embedding and utilize ViT for learning image-modality representations. **-BERT+ResNet**: Following Pan et al. (2020), we replace the ViT in our framework with ResNet-152 (He et al., 2016) to embed each image patch as a 2048-dimensional vector. **-GloVe+ResNet**: We use GloVe to acquire word embeddings and employ ResNet for learning image-modality representations.

A.3 Proof of Proposition 1

Proof. The proof of mutual information’s lower bound estimation can be found in the appendix of Oord et al. (2018). Here we present the proof for the upper bound estimation. We know that the KL divergence is always greater than zero, and therefore we have:

$$D_{KL}(p(\mathbf{x})||q(\mathbf{x})) = \mathbb{E}_{p(\mathbf{x})}[\log p(\mathbf{x})] - \mathbb{E}_{q(\mathbf{x})}[\log q(\mathbf{x})] \geq 0. \quad (12)$$

By following the definition of mutual information, we get:

$$\begin{aligned} \mathcal{I}(\mathbf{x}; \mathbf{y}) &= \mathbb{E}_{p(\mathbf{x}, \mathbf{y})} \left[\log \frac{p(\mathbf{x}|\mathbf{y})}{p(\mathbf{x})} \right] \\ &\approx \mathbb{E}_{p(\mathbf{x}|\mathbf{y})} \left[\log \frac{p(\mathbf{x}|\mathbf{y})}{p(\mathbf{x})} \right] \\ &\leq \mathbb{E}_{p(\mathbf{x}|\mathbf{y})} \left[\log \frac{p(\mathbf{x}|\mathbf{y})}{p(\mathbf{x}|\mathbf{y})} \right] \\ &= D_{KL}(p(\mathbf{x}|\mathbf{y})||q(\mathbf{x})). \end{aligned} \quad (13)$$

Thus, we conclude:

$$\mathcal{I}(\mathbf{x}; \mathbf{y}) \leq D_{KL}(p(\mathbf{x}|\mathbf{y})||q(\mathbf{x})). \quad (14)$$

□